

Being an open source project, the platform encourages community contributions, innovation, and the development of compatible tools and extensions. This fosters a collaborative ecosystem where developers can build upon and customise Mozilla Hubs to suit their specific needs.

Other open source metaverse platforms

Apart from JanusVR, there are several other open source metaverse platforms available in 2023 that provide diverse and immersive virtual reality experiences. These platforms offer unique features, innovative technologies, and decentralised environments, allowing users and developers to explore and create in the metaverse. Here are the top options.

Webaverse: Webaverse is a decentralised VR platform with top technical features. It utilises blockchain technology for asset ownership and secure transactions, allowing users to create, buy, sell, and trade virtual goods using cryptocurrencies and NFTs. Webaverse emphasises decentralisation, hosting virtual worlds on distributed peer-to-peer networks. Users can explore interactive virtual environments, engage in multiplayer games, and collaborate on creative projects. The platform supports user-generated content and embraces an open source approach, encouraging community participation and innovation within the Webaverse ecosystem.

Hypercube: Hypercube is a virtual reality (VR) platform that offers immersive and interactive experiences. It enables users to navigate and interact with virtual environments using advanced VR technologies. With its spatial computing capabilities, Hypercube allows users to move freely within virtual spaces, providing a sense of presence and realism. The platform supports various input methods, including hand gestures and motion controllers, enhancing user engagement. Hypercube's aim is to provide a

seamless and immersive VR experience that blurs the line between the physical and virtual worlds.

Somnium Space: Somnium Space is a virtual reality (VR) platform that offers users a persistent, immersive, and user-generated metaverse. Users can explore and interact with a vast virtual world filled with unique landscapes, buildings, and user-created content. Somnium Space emphasises blockchain technology, enabling true ownership of virtual land and assets through non-fungible tokens (NFTs). Users can socialise, attend events, trade virtual goods, and even build and monetise their own experiences.

XREngine: XREngine is a cutting-edge game engine designed for creating immersive virtual reality (VR) experiences. Built from the ground up for VR, XREngine offers advanced features and optimisations specifically tailored to the unique demands of VR development. It supports high-performance rendering, physics simulations, and spatial audio, providing a realistic and immersive environment. XREngine also includes tools and frameworks for creating interactive gameplay mechanics and immersive storytelling. Its goal is to empower

developers to create captivating VR experiences with seamless performance and visual fidelity.

WebXR Device API: The WebXR Device API is a web standard that provides a unified and consistent way to access and interact with virtual reality (VR) and augmented reality (AR) devices through web browsers. It allows developers to create immersive and interactive XR experiences that can be accessed directly from a browser, without the need for additional plugins or applications. The API provides capabilities such as tracking the user's head and hand movements, rendering VR and AR content, and integrating with input devices like controllers. It aims to democratise XR development and make XR experiences easily accessible on the web.

Open source platforms allow users and developers to shape the metaverse according to their needs and preferences, leading to a diverse and dynamic virtual reality landscape. As the technology progresses and more platforms emerge, the metaverse is poised to revolutionise various industries, including education, entertainment, commerce, and healthcare. **END** 

References

- <https://www.linux.com/learn/exploring-open-source-virtual-reality-development-platforms>
- <http://www.timdwyer.net/opensim/>
- <https://www.linuxjournal.com/content/exploring-open-source-virtual-reality-platforms>
- <https://openmetaverse.org>
- http://opensimulator.org/wiki/Main_Page
- <http://www.janusvr.com>
- <https://hubs.mozilla.com>
- <https://www.kitely.com>
- <https://metaverseofthing.com/top-10/open-source-metaverse-platforms-and-projects/>

By: Dr Magesh Kasthuri and Dr Anand Nayyar

Dr Magesh Kasthuri is a senior distinguished member of the technical staff and principal consultant at Wipro Ltd. This article expresses his views and not that of Wipro.

Dr Anand Nayyar is a PhD in wireless sensor networks and swarms intelligence. He works at Duy Tan University, Vietnam, and loves to explore open source technologies, IoT, cloud computing, deep learning and cyber security.